

Basic LLM Chatbot Using Gemini

1. Title :-

Basic LLM Chatbot Using Google Gemini API

2. Aim :-

To create a simple Large Language Model (LLM) application using the **Google Gemini API** in Google Colab and generate answers for user questions.

3. Objective :-

- To understand the basic concept of an LLM.
- To connect Python with the Gemini API.
- To use an API key securely through Google Colab Secrets.
- To send a question to the Gemini model.
- To display the generated answer.

4. Software and Tools Used

Tool	Purpose
Google Colab	To write and execute Python code
Python	Programming language
Google GenAI	To communicate with Gemini
LangChain	LLM application framework/library
Gemini 3.6 Flash	LLM used for generating responses
Gemini API Key	To access the Gemini API

5. Procedure:-

Step 1: Install Required Packages

```
[ ] | pip install -q -U google-genai langchain
```

Step 2:- Import Required Libraries.

```
[ ] from google import genai
    from google.colab import userdata
```

Step 3 : Get the API Key

```
[ ] key = userdata.get("GeminiAPIKey")
    print("API key loaded successfully")
    API key loaded successfully
```

Step 4 : Initialize the Gemini Client

```
[ ] client = genai.Client(api_key=key)
```

Step 5 : Define the Prompt

```
[ ] prompt_text = "What is Cloud Computing?"
```

Step 6 : Generate the Response

```
[ ] response = client.interactions.create(
    model="gemini-3.6-flash",
    input=prompt_text
)
```

Step 7 : Display the Response

```
[ ] print(response.output_text)

... At its simplest, cloud computing is the delivery of computing services—including servers, storage, databases, networking, software, and
    Instead of buying, owning, and maintaining physical data centers or servers, you rent access to technology services on an as-needed basis f
    ---
    ### An Everyday Analogy
    Think of cloud computing like electricity in your home:
    * You don't build your own power plant to light your house.
    * Instead, you plug into the power grid, use as much electricity as you need, and pay only for what you consume.
    * Cloud computing works the same way, but with computing power and data storage instead of electricity.
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    ### Examples You Probably Use Every Day
    If you use an online service to send email, edit documents, watch movies, listen to music, or store pictures, cloud computing is likely mak
    * Streaming Services: Netflix, Spotify, Disney+
    * Cloud Storage: Google Drive, Apple iCloud, Dropbox
    * Productivity Tools: Microsoft 365, Google Docs, Zoom
```

Conclusion:-

The assignment was successfully completed by creating a basic LLM application using **Google Gemini 3.6 Flash** and the Google GenAI API in Google Colab. The program takes a user question as input, sends it to the LLM, and displays the generated response.